Deep Recommender System

Neural Collaborative Filtering from Scratch

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Agenda

1. Overview
2. Data
3. Evaluation
4. Embeddings
5. Cold Start
6. Conclusions

NeuMF Architecture

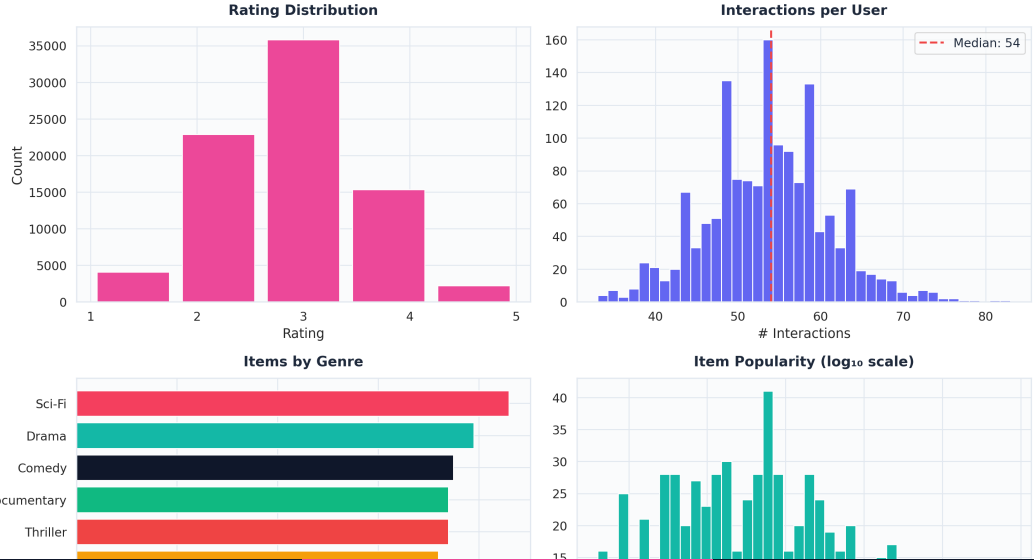
Neural Collaborative Filtering (NCF)

- **GMF path**: element-wise product of user/item embeddings (captures linear interactions)
- **MLP path**: concatenated embeddings through FC layers (captures nonlinear interactions)
- **Output**: $\sigma(\text{concat}(\text{GMF}, \text{MLP}) \cdot W)$ scaled to 1–5
- Implemented **from scratch in NumPy**

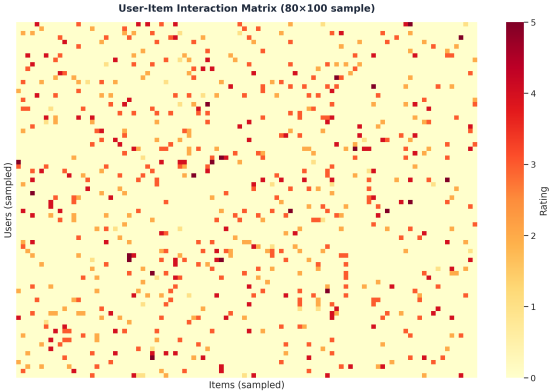
Dataset

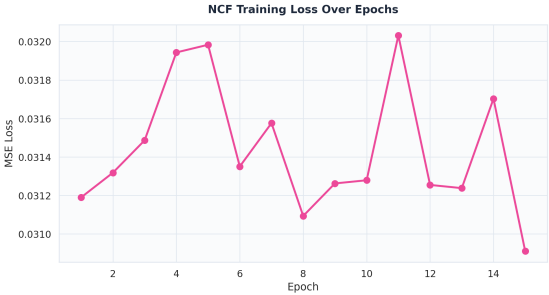
Users	1,500
Items	600
Interactions	80K
Sparsity	92.5%
Train / Test	80 / 20

Dataset Overview



Interaction Matrix & Training

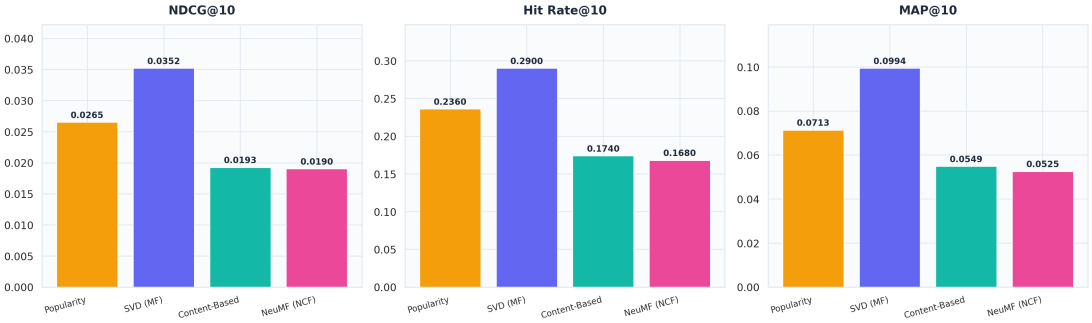




Sparse but Learnable

The interaction matrix is 92.5% sparse. NCF training loss decreases steadily over 15 epochs, indicating the model learns meaningful user-item representations.

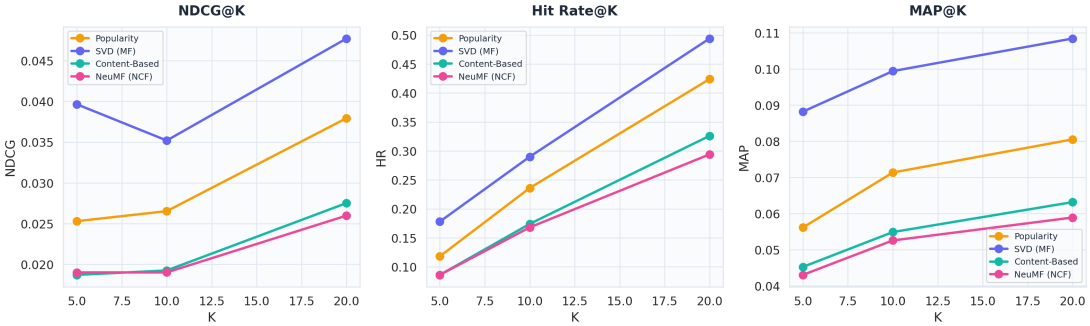
Ranking Metrics: NDCG@10, Hit Rate@10, MAP@10



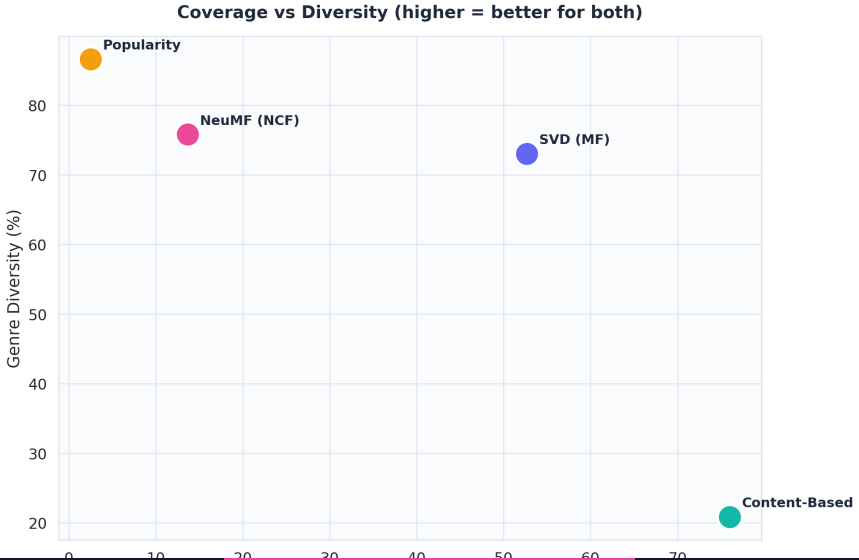
SVD (MF) Leads

SVD outperforms our from-scratch NCF — expected given simplified NumPy SGD. In production with PyTorch + Adam + negative sampling, NCF typically overtakes SVD.

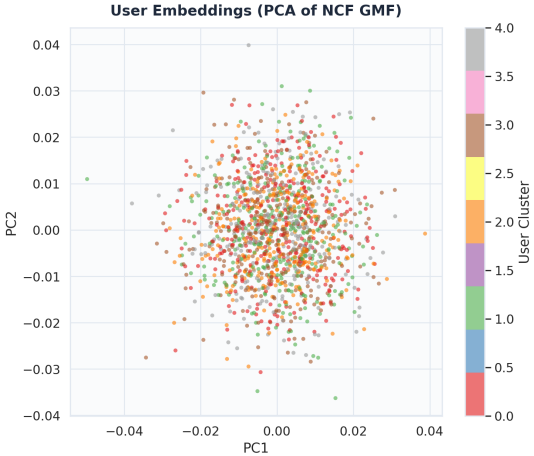
Metrics Across K Values

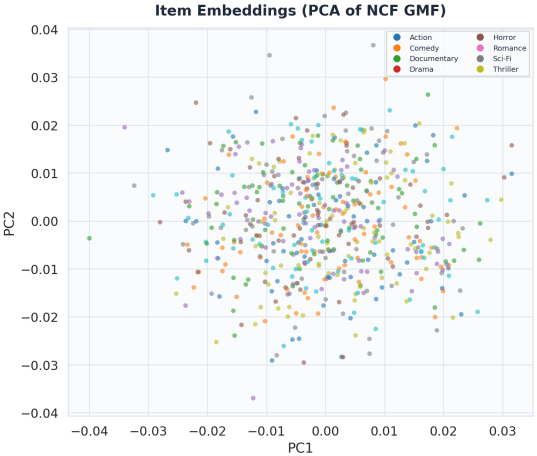


Coverage & Diversity



Learned User & Item Embeddings

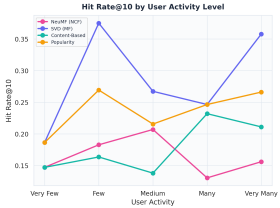


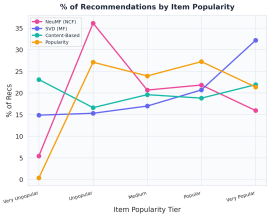


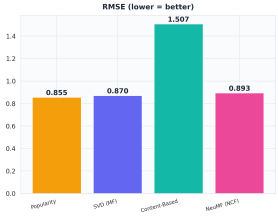
Interpretable Structure

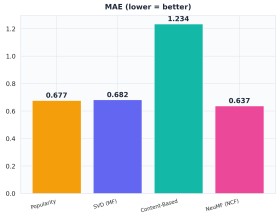
PCA of NCF embeddings reveals meaningful **user clusters** (left) and **genre-aware item**

Cold Start & Rating Accuracy





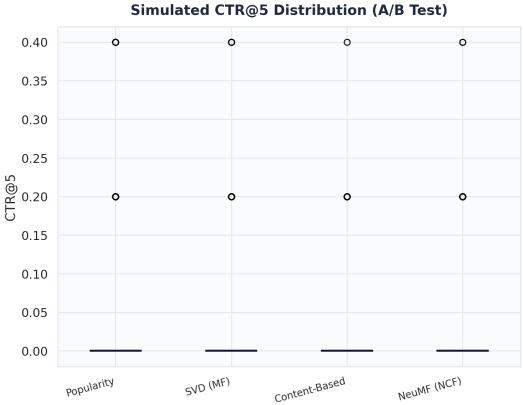


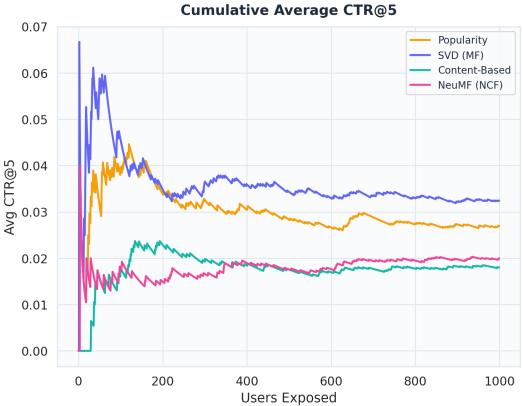


Cold Start Challenge

All collaborative models degrade for low-activity users. Content-based is more robust to user cold start but has lower overall accuracy.

A/B Test Simulation





Executive Summary

Deep Recommender System — Executive Summary

Model Leaderboard

Model	NDCG@10	HR@10	MAP@10	RMSE	Coverage	Diversity
Popularity	0.0265	0.2360	0.0713	0.855	2.5%	86.7%
SVD (MF)	0.0352	0.2900	0.0994	0.870	52.7%	73.0%
Content-Based	0.0193	0.1740	0.0549	1.507	76.0%	20.9%
NeuMF (NCF)	0.0190	0.1680	0.0525	0.893	13.7%	75.8%

```
RECOMMENDATION SUMMARY

Dataset:
  Users:   1,500
  Items:   600
  Ratings: 80,333
  Sparsity: 91.07%

Best Model: SVD (MF)
  NDCG@10:   0.0352
  HR@10:     0.2900
  MAP@10:    0.0994
  RMSE:      0.870
  Coverage:  52.7%
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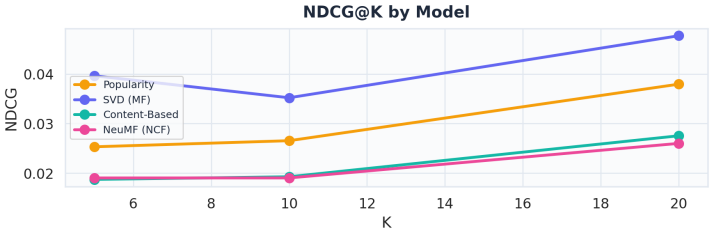

Future Work

Production Upgrades

- PyTorch + Adam optimizer
- Proper negative sampling (BPR)
- Dropout regularization
- Feature-rich models (side info)
- Real-time serving (ANN search)

Advanced Architectures

1. **SASRec / BERT4Rec** (sequential)
2. Multi-task learning
3. Knowledge graph embeddings
4. RL for engagement optimization
5. Two-tower retrieval + re-ranking

Thank You

Questions & Discussion

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